

ROSHAN FRANCIS

Computer Science Undergraduate | Backend & AI Systems

Thiruvananthapuram, Kerala | +91 9446646004 | fbiju213@gmail.com | github.com/Roshan-Francis

SUMMARY

Computer Science undergraduate (CGPA 8.26) with hands-on experience building backend services and AI-powered APIs in Python (FastAPI, Flask), alongside Flutter mobile apps and ESP32/LoRa IoT systems. Built and deployed a Dockerized resume-scoring microservice using semantic embeddings and a face-recognition attendance system across internship, academic, and team projects, with a focus on clean API design and reliable data storage.

TECHNICAL SKILLS

Languages: Python (Intermediate), Java (Beginner), C (Beginner), SQL (Beginner)

Core CS: Data Structures & Algorithms, OOP, DBMS, REST API Design

Frameworks & Libraries: FastAPI, Flask, Flutter, Java Swing, OpenCV, sentence-transformers

Databases: PostgreSQL, Supabase, MySQL, Firebase Firestore

Tools & Platforms: Docker, Git, Hugging Face Spaces, Streamlit

EXPERIENCE

Assistive Technology Intern — IEEE EMBS Kerala Chapter & MBCET

Jun 2025 | Thiruvananthapuram, Kerala

- Completed an 11-day internship studying how technology can assist people with physical and cognitive disabilities.
- Designed and built prototype IR-sensor glasses to help children with autism manage personal-space awareness, triggering a red-light alert when someone crosses a set proximity boundary.

Flutter App Development Intern — ICT Academy of Kerala

Dec 2024 – Jan 2025 | Thiruvananthapuram, Kerala

- Completed a one-month intensive program building cross-platform mobile apps with Flutter and Dart, covering UI design and state management.
- Built a Hangman word game app in a 4-member team, with random word selection, animations, and persistent high scores using Shared Preferences (github.com/Roshan-Francis/hangman).

FEATURED PROJECTS

Resume Scoring API — AI-Powered Resume Screening Microservice

github.com/Roshan-Francis/Resume_checker_api | Live: roshfr-resume-v2.hf.space

- Built a FastAPI `/score-resume` endpoint that scores a resume PDF against a job description on a 0–100 scale, returning a match label, detected CGPA, CGPA-threshold warnings, and personalized improvement tips.
- Designed a weighted scoring model: keyword match with synonym resolution (30), semantic similarity via Qwen3-Embedding-0.6B cosine similarity (30), experience signals (25), and achievements (15).
- Containerized with Docker and deployed on Hugging Face Spaces with live Swagger docs; built a Streamlit frontend for testing.

HOWL — Group Motorcycle Rider Safety System

Backend Developer | Ongoing

- Developing the backend for an IoT rider-safety system (Flutter app + ESP32/LoRa hardware with an offline LoRa mesh fallback), building multiple FastAPI REST APIs that power the mobile app.
- Designed the Supabase (PostgreSQL) database and implemented Google login through Supabase Auth.
- Integrated Mapbox to deliver the app's map and location features.

SwiftVision — Face Recognition Attendance System

Feb 2025 – Apr 2025 | github.com/Roshan-Francis/swiftvision

- Built a Flutter app with Google Sign-In (Firebase Auth) to register students and mark attendance from a photo.
- Developed a Flask backend that detects faces with MTCNN/OpenCV, matches them against registered student images, and records attendance in both MySQL and Firestore for redundancy.

Student Result Management System

github.com/Roshan-Francis/java-project

- Built a Java Swing desktop app backed by MySQL for teachers to enter and manage marks, and for students to view results with pass/fail status.

ACHIEVEMENTS

- **National Qualifier — Smart India Hackathon (SIH) 2025:** Cleared the internal college round as Team Hackonauts, advancing to the national-level hackathon pool.

EDUCATION

B.Tech in Computer Science & Engineering — Mar Baselios College of Engineering and Technology (Autonomous)

CGPA: 8.26 | 2023 – 2027

Class XII (CBSE): 78% (2023) | Class X (CBSE): 89.6% (2021) — Shanthinikethan School, Thiruvananthapuram

CERTIFICATIONS

- **National Bootcamp on IoT Forensics — C-DAC & MBCET (ISEA Project, MeitY), Dec 2025:** device data extraction, packet analysis, hex-level data recovery, and IoT security analysis.